

IMO(30th) – 1989

First Day

1. Prove that the set $\{1, 2, \dots, 1989\}$ can be expressed as the disjoint union of 17 subsets $A_1, A_2, \dots, A_{17}$ such that:
 - (a) Each A_i contains the same number of elements.
 - (b) The sum of all elements of A_i is the same for $i = 1, 2, \dots, 17$.
2. Let ABC be a triangle. The bisector of angle A meets the circumcircle of triangle ABC in A_1 . Points B_1 and C_1 are defined similarly. Let AA_1 meet the lines that bisect the two external angles at B and C in point A^0 . Define B^0 and C^0 similarly. If $S_{x_1 x_2 \dots x_n}$ denotes the area of the polygon $x_1 x_2 \dots x_n$, prove that $S_{A^0 B^0 C^0} = 2S_{AC_1 B A_1 C B_1} \geq 4S_{ABC}$.
3. Given a set S in the plane containing n points and satisfying the conditions:
 - (a) No three points of S are collinear.
 - (b) For every point P of S there exist at least k points in S that have the same distance to P .

Prove that the following inequality holds: $k < \frac{1}{2} + \sqrt{2n}$.

Second Day

4. The quadrilateral $ABCD$ has the following properties:
 - (a) $AB = AD + BC$
 - (b) There is a point P inside it at a distance x from the side CD such that $AP = x + AD$ and $BP = x + BC$.
 Show that $\frac{1}{\sqrt{x}} \geq \frac{1}{\sqrt{AD}} + \frac{1}{\sqrt{BC}}$.
5. For which positive integer n does there exist a positive integer N such that none of the integers $1 + N, 2 + N, \dots, n + N$ is the power of a prime number?
6. We consider permutations $(x_1, x_2, \dots, x_{2n})$ of the set $\{1, 2, \dots, 2n\}$ such that $|x_i - x_{i+1}| = n$ for at least one $i \in \{1, 2, \dots, 2n - 1\}$. For every natural number n , find out whether permutations with this property are more or less numerous than the remaining permutations of $\{1, 2, \dots, 2n\}$.

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